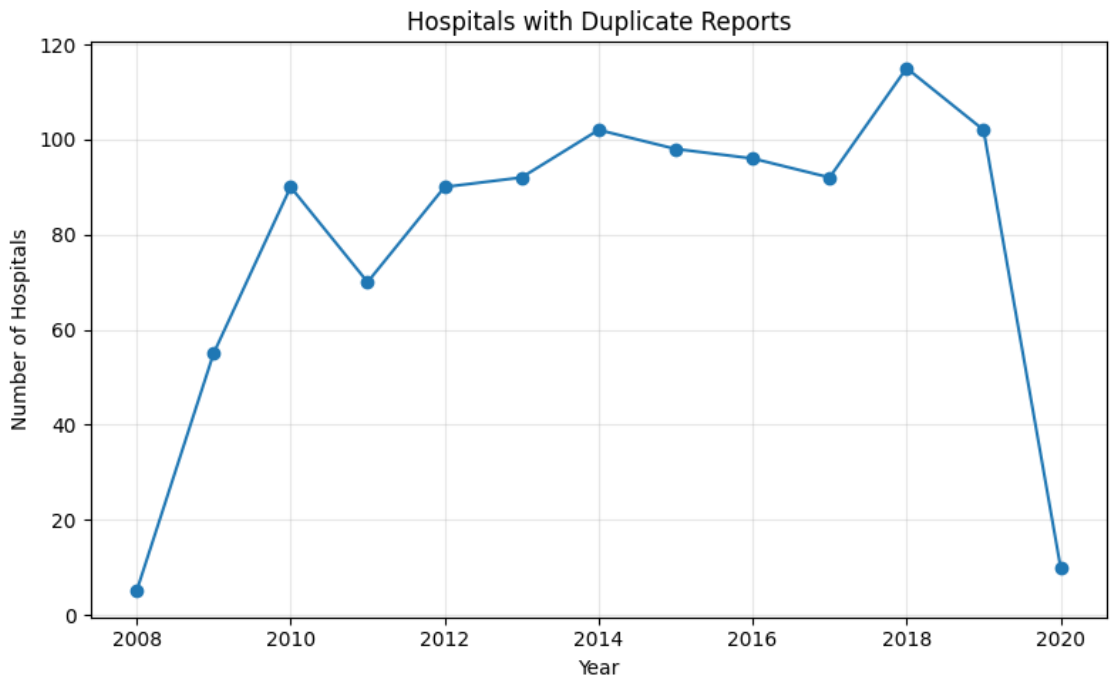
Danzilo_G_HW_4_3

April 3, 2026

0.1 Question 1

	fyear	provider_number	num_reports
106	2008	30061	2
118	2008	30093	2
182	2008	40151	2
259	2008	50158	2
271	2008	50230	2

	fyear	num_hospitals
0	2008	42
1	2011	45
2	2012	90
3	2013	91
4	2014	102
5	2015	96
6	2016	95
7	2017	91
8	2018	101

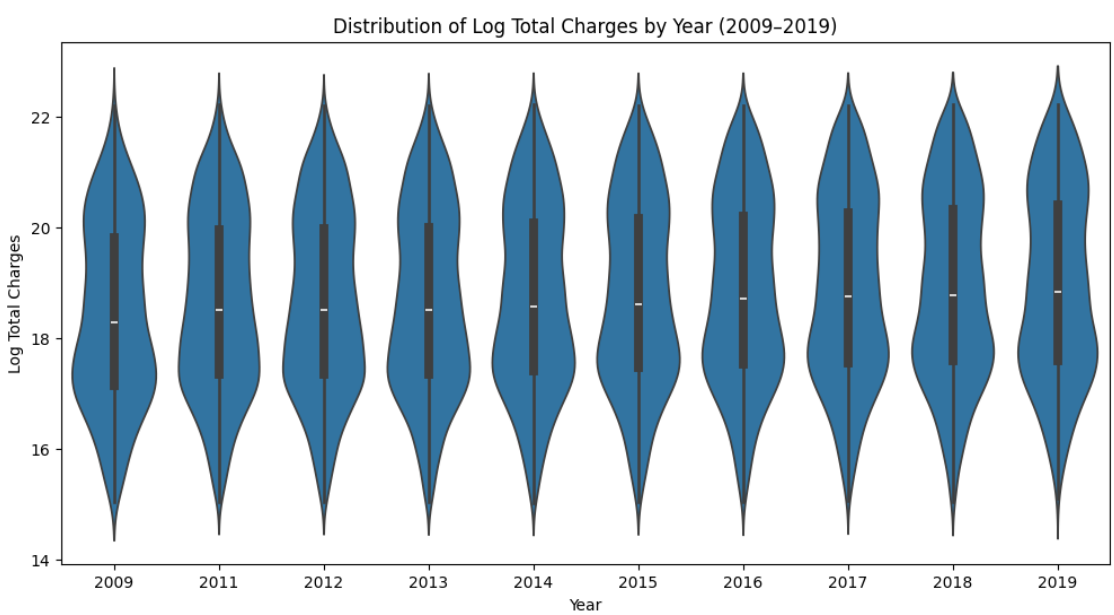


0.2 Question 2

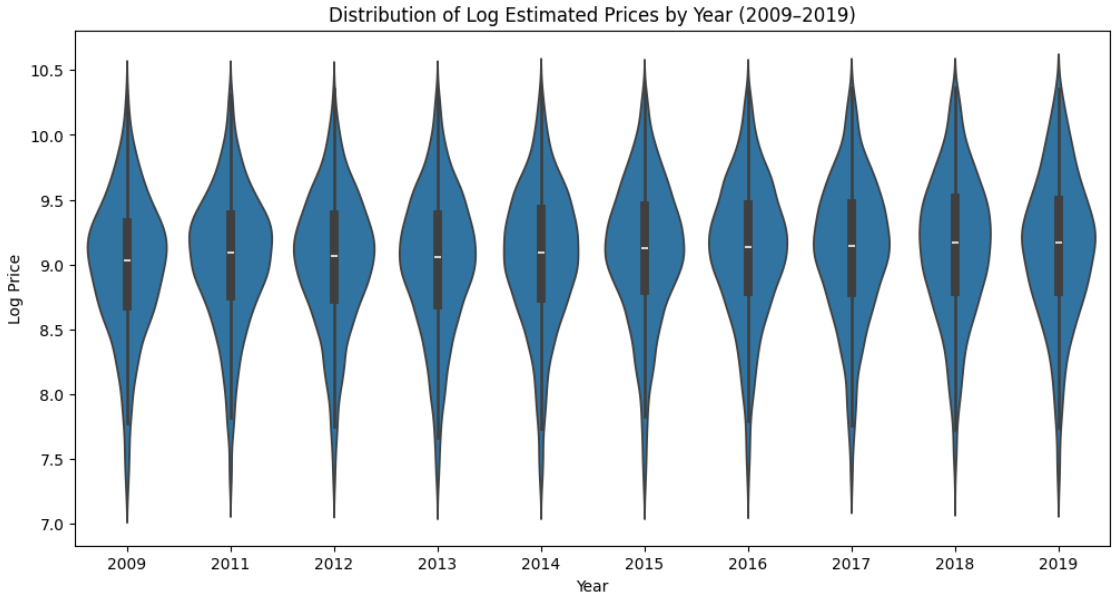
6732

There are 6793 unique hospital IDs.

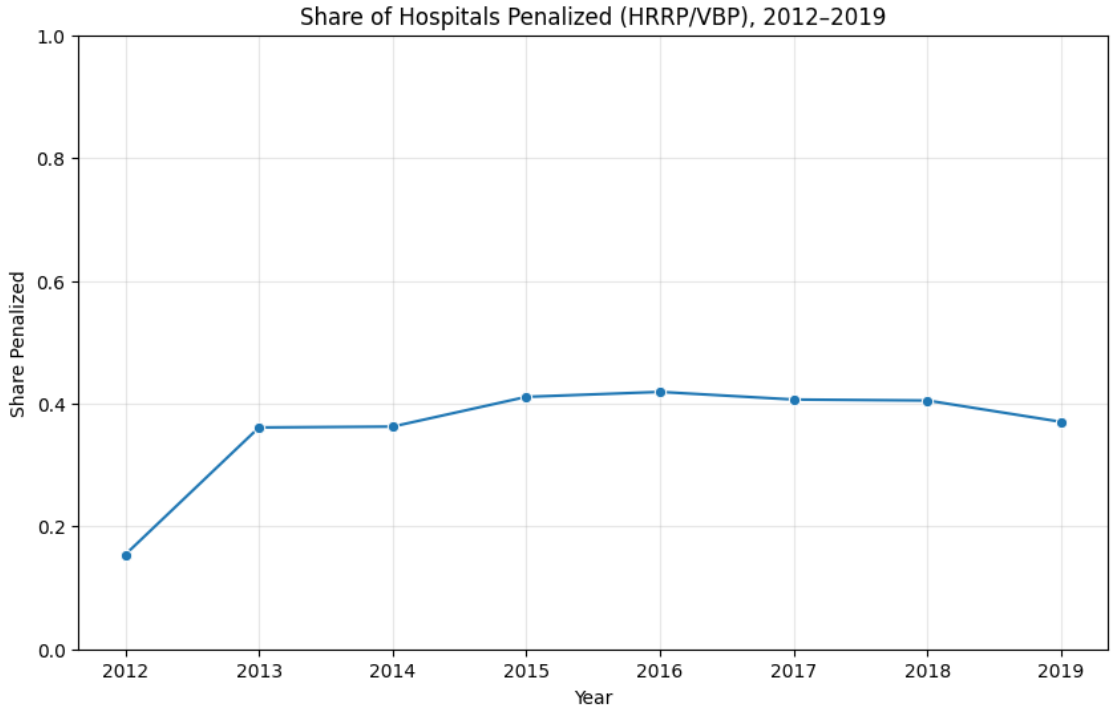
0.3 Question 3



0.4 Question 4



0.5 Question 5



0.6 Question 6

```
N in analysis: 5557

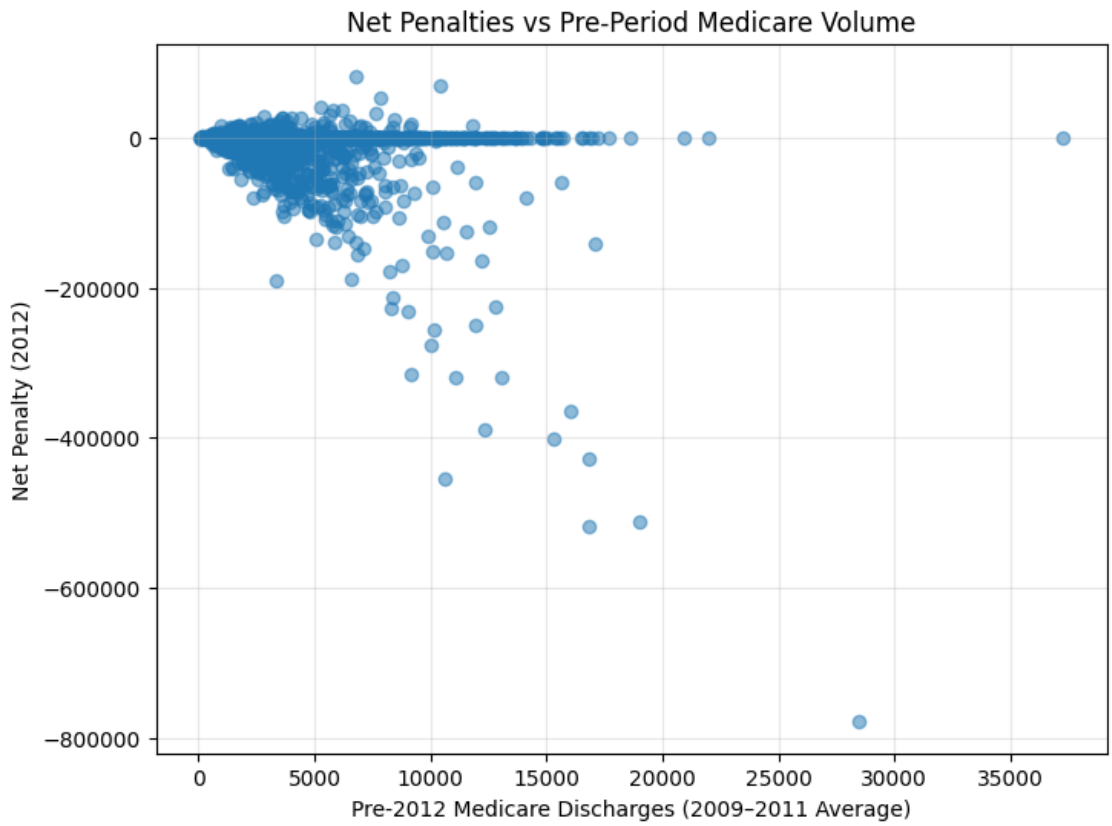
price_2011: 5815
price_2014: 6064
pen_2012: 6140
pre_means: 6169
dup in price_2011: 0
dup in price_2014: 0
dup in pen_2012: 0
dup in pre_means: 0

N in analysis: 2376
price_2011: 2621
```

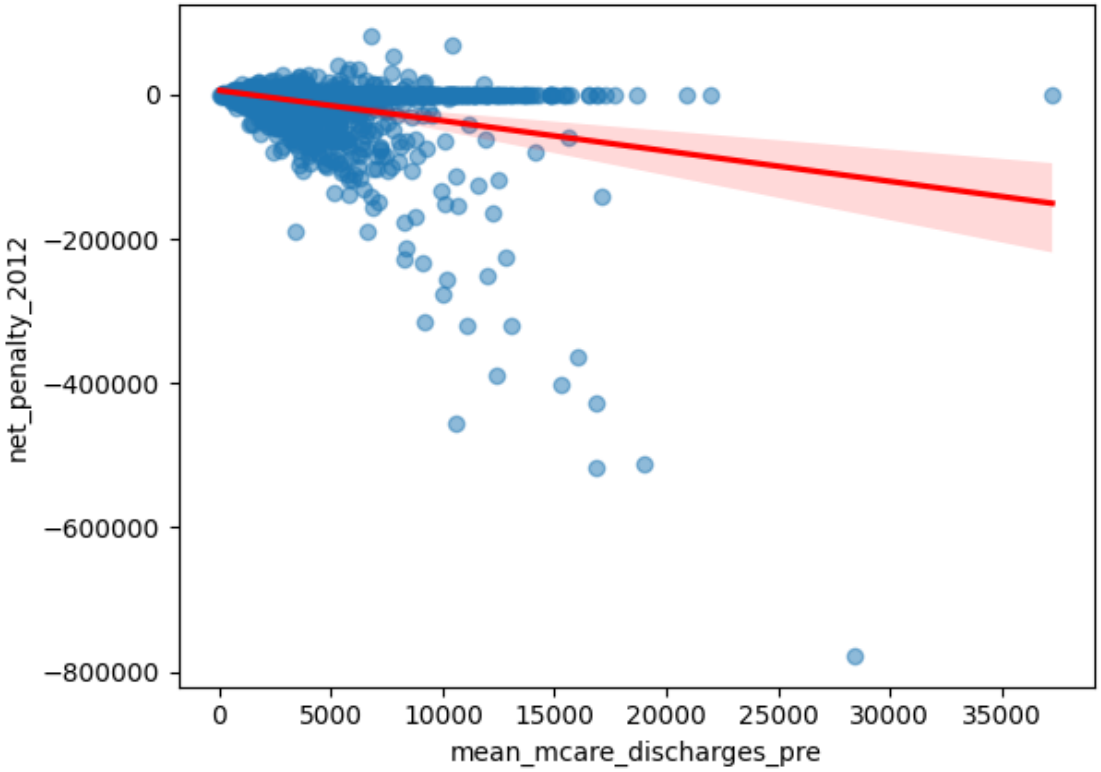
	(1)	(2)	(3)
Intercept	771.3117*	-190.6956	-209.0165
	(406.8079)	(385.2290)	(406.9617)
net_penalty_2012	-0.0027	0.0045	0.0060
	(0.0028)	(0.0079)	(0.0100)
mean_beds_pre		4.6235	6.5891
		(3.6151)	(6.2565)
mean_mcaid_discharges_pre			-0.2501
			(0.3511)
R-squared	0.0000	0.0021	0.0025
R-squared Adj.	-0.0004	0.0012	0.0012
N	2358	2358	2358
R2	0.000	0.002	0.002

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

0.7 Question 7




```
<Axes: xlabel='mean_mcare_discharges_pre', ylabel='net_penalty_2012'>
```



0.8 Question 8

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==============================================================================
===============
                              FS(1)       FS(2)       FS(3)       RF(1)
RF(2)      RF(3)
------------------------------------------------------------------------------
---------------
Intercept                  5648.7735** 6348.2367** 6265.5726** -1039.8389
-1094.2949 -250.0924
                           (2864.3257) (3187.4267) (3111.3976) (1032.2865)
(990.8787) (516.1313)
mean_mcare_discharges_pre -4.1952***  -2.9940***  -2.3623***  0.3993      0.3058
0.1736
                           (0.9378)    (0.8487)    (0.8470)    (0.2944)
(0.4640)   (0.4107)
mean_beds_pre                          -22.8178    -54.4047**              1.7765
3.2626*
                                       (17.2659)   (24.1793)
(3.8308)   (1.7282)
mean_mcaid_discharges_pre                          2.9764***
```

```
-0.1951
                                                    (0.9388)
(0.2407)
R-squared                 0.1130      0.1159      0.1295      0.0024      0.0024
0.0026
R-squared Adj.            0.1126      0.1151      0.1284      0.0020      0.0016
0.0013
N                         2376        2376        2358        2376        2376
2358
R2                        0.113       0.116       0.129       0.002       0.002
0.003
================================================================================
===============
Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01
```

0.9 Question 9*

```
                          IV-2SLS Estimation Summary
==============================================================================
Dep. Variable:           price_change   R-squared:                     -0.0183
Estimator:                    IV-2SLS   Adj. R-squared:                -0.0188
No. Observations:                2358   F-statistic:                    1.0208
Date:                Fri, Apr 03 2026   P-value (F-stat)                0.3123
Time:                        01:46:28   Distribution:                  chi2(1)
Cov. Estimator:                robust

                                Parameter Estimates
================================================================================
====
                  Parameter  Std. Err.     T-stat    P-value    Lower CI
Upper CI
--------------------------------------------------------------------------------
----
Intercept            155.27     295.53     0.5254     0.5993     -423.96
734.49
net_penalty_2012    -0.0662     0.0655    -1.0103     0.3123     -0.1946
0.0622
================================================================================
====

Endogenous: net_penalty_2012
Instruments: mean_mcare_discharges_pre
Robust Covariance (Heteroskedastic)
Debiased: False
                          IV-2SLS Estimation Summary
==============================================================================
Dep. Variable:           price_change   R-squared:                     -0.0217
```

```
Estimator:                    IV-2SLS   Adj. R-squared:                -0.0226
No. Observations:                2358   F-statistic:                    4.1995
Date:                Fri, Apr 03 2026   P-value (F-stat)                0.1225
Time:                        01:46:28   Distribution:                  chi2(2)
Cov. Estimator:                robust

                              Parameter Estimates
================================================================================
====
                  Parameter  Std. Err.     T-stat    P-value    Lower CI
Upper CI
--------------------------------------------------------------------------------
----
Intercept            200.93     569.58     0.3528     0.7243     -915.43
1317.3
mean_beds_pre       -0.4498     7.4536    -0.0603     0.9519     -15.059
14.159
net_penalty_2012    -0.0718     0.1547    -0.4643     0.6424     -0.3751
0.2314
================================================================================
====

Endogenous: net_penalty_2012
Instruments: mean_mcare_discharges_pre
Robust Covariance (Heteroskedastic)
Debiased: False
                          IV-2SLS Estimation Summary
==============================================================================
Dep. Variable:           price_change   R-squared:                     -0.0227
Estimator:                    IV-2SLS   Adj. R-squared:                -0.0240
No. Observations:                2358   F-statistic:                    4.1729
Date:                Fri, Apr 03 2026   P-value (F-stat)                0.2434
Time:                        01:46:28   Distribution:                  chi2(3)
Cov. Estimator:                robust

                                   Parameter Estimates
================================================================================
=============
                           Parameter  Std. Err.     T-stat    P-value    Lower
CI    Upper CI
--------------------------------------------------------------------------------
-------------
Intercept                     210.44     683.36     0.3080     0.7581
-1128.9      1549.8
mean_beds_pre                -0.7363     11.155    -0.0660     0.9474
-22.600      21.127
mean_mcaid_discharges_pre     0.0236     0.3202     0.0738     0.9412
-0.6039      0.6512
```

```
net_penalty_2012             -0.0735     0.1757    -0.4183     0.6757
-0.4179      0.2709
================================================================================
=============

Endogenous: net_penalty_2012
Instruments: mean_mcare_discharges_pre
Robust Covariance (Heteroskedastic)
Debiased: False
```

	(1)	(2)	(3)
Intercept	155.267	200.931	210.440
Intercept SE	295.529	569.581	683.358
Net Penalty (IV)	-0.066	-0.072	-0.074
Net Penalty SE	0.066	0.155	0.176
Avg Beds (Pre-2012)	NaN	-0.450	-0.736
Beds SE	NaN	7.454	11.155
Avg Medicaid Discharges (Pre-2012)	NaN	NaN	0.024
Medicaid SE	NaN	NaN	0.320
Num.Obs.	2358.000	2358.000	2358.000
R2	-0.018	-0.022	-0.023

0.10 Question 10

The IV estimate identifies a local ATE, reflecting the effect of penalties on prices for hospitals whose penalty status is influenced affected by their pre-2012 Medicare discharges. This effect applies mostly to hospitals with higher Medicare exposure and can be different from overall ATE, reflecing the average effect across all hospitals.